

Credible on Paper? Green Banking Policy, Access to Finance, and Institutional Capacity in the Greening of Firms Worldwide

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Abstract

Green banking policies, coordinated through the Sustainable Banking and Finance Network (SBFN) across sixty-plus emerging economies, assume that supervisory guidelines directing banks to price environmental risk strengthen the link between firms' credit access and green investment. Using two independent World Bank Enterprise Survey samples (28,042 firms, 41 economies; 89,797 firms, 162 economies) and four estimators—pooled logit, Bayesian hierarchical, causal forest, and country fixed effects—we find bank credit access robustly predicts green-practice adoption (13 percentage points), but neither SBFN adoption nor regulatory quality moderates this relationship. The estimates are precise, not merely insignificant: the causal forest's credit effect is 0.072 for non-adopter firms versus 0.076 for adopter firms, and is nearly identical across regulatory-quality terciles, ruling out moderation of policy-relevant size. A country-level staggered event-study (217 economies)

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corroborates this at the macro level. Heterogeneity instead concentrates at firm size, relocating rather than resolving which institutional margin matters.

Keywords: green banking policy, access to finance, institutional complementarity, firm-level green investment, causal forest, hierarchical Bayesian model

JEL: G21, O16, Q56, D22, C11

1. Introduction

Since 2012, more than sixty central banks and financial regulators across the emerging and transition world have signed on to a coordinated effort to green their financial systems through the IFC’s Sustainable Banking and Finance Network (SBFN), directing commercial banks to price environmental risk into lending and channel credit toward cleaner production. Whether that adoption changes what happens on the ground—whether a firm that can get a bank loan is any more likely to install pollution-control equipment or measure its own emissions—is a separate and harder question, and it is the one this paper answers, with enough statistical power to distinguish a true absence of effect from merely an underpowered test of one.

A large literature on state capacity argues that a formally identical policy can produce different real-economy outcomes depending on the administrative infrastructure behind it (La Porta et al., 1998; Acemoglu et al., 2001), and recent country-level evidence finds green-finance policy effectiveness itself conditional on governance quality (Jia et al., 2026; Chi and Yang, 2023; Shi et al., 2024; W. Ullah et al., 2024). Each of these results implies a firm-level prediction—that the credit-to-green channel should respond to policy more

19 sharply where regulatory capacity is stronger—that has not been tested di-
 20 rectly, across a broad cross-section of adopting economies, on firm-level data.
 21 A separate strand shows firm-level financing constraints shape green invest-
 22 ment (Ghosh and Dutta, 2022; Costa et al., 2024; De Haas et al., 2025) and
 23 that the WBES Green Economy module is a workable lens on firm environ-
 24 mental behavior (Asif et al., 2025; Phan, 2026), but nothing yet connects the
 25 two literatures at firm level.

26 We test this using two independently assembled World Bank Enterprise
 27 Survey (WBES) samples: a 41-economy primary sample (28,042 firms, 2018–
 28 2023) carrying the Bank’s full Green Economy module, and a 162-economy
 29 global sample (89,797–100,115 firms, 2021–2026) reaching, for the first time
 30 in this literature, into high-income economies entirely outside SBFN’s remit
 31 (Section 4). We match each economy to its SBFN policy status as of its sur-
 32 vey year and to the World Bank’s Worldwide Governance Indicators. Because
 33 SBFN status is fixed within country in a single wave, a country fixed effect
 34 absorbs its main effect entirely; its interaction with credit access remains
 35 formally identified, but off only the eight adopter economies in the primary
 36 sample, which is why we also estimate a Bayesian hierarchical model with
 37 country-varying slopes (Gelman and Hill, 2007; Gelman et al., 2020) and a
 38 causal-forest / double-machine-learning estimator (Wager and Athey, 2018;
 39 Chernozhukov et al., 2018; Athey et al., 2019; Athey and Imbens, 2019) that
 40 let the data reveal moderation rather than relying on a single fragile inter-
 41 action term. On the larger global sample, country fixed effects are usable
 42 directly. Across all four estimators the answer is the same, and precisely so:
 43 the causal forest’s credit effect differs by only 0.004 between SBFN adopters

44 and non-adopters, and is flat across regulatory-quality terciles—these are
45 tight estimates that rule out policy-relevant moderation, not underpowered
46 nulls that merely fail to detect one.

47 This paper contributes in five respects. First, it is, to our knowledge,
48 the first direct firm-level test of whether SBFN-style adoption changes the
49 credit-to-green channel, complementing the bank-level, supply-side evidence
50 that dominates this literature. Second, an independently assembled 162-
51 economy global sample replicates the primary sample’s result under a mate-
52 rially sharper identification strategy, a different outcome measure, and a set of
53 economies sharing almost no overlap with the primary sample—considerably
54 more informative than either sample alone. Third, because our estimates
55 are precise rather than merely insignificant, we can distinguish a genuine
56 absence of institutional moderation from a design too underpowered to de-
57 tect one, a distinction most single-country studies in this literature cannot
58 make. Fourth, having ruled out country-institutional moderation across four
59 estimators and two samples, we show (Section 3.4) that the heterogeneity
60 that does exist concentrates at firm size rather than country institutions,
61 relocating rather than resolving the question of which margin governs pol-
62 icy effectiveness—reported throughout as an exploratory, non-pre-registered
63 finding rather than a settled result. Fifth, because none of the above exploits
64 SBFN’s own adoption timing, Section 6 builds an independent country-year
65 panel and estimates a Callaway-Sant’Anna staggered event-study—the de-
66 sign class this literature most often asks for—which likewise finds no benefi-
67 cial institutional effect on aggregate environmental outcomes.

68 Section 2 sets out SBFN’s institutional background; Section 3 develops

69 the paper’s hypotheses; Section 4 describes data and empirical strategy; Sec-
70 tion 5 presents results; Section 7 discusses implications; and Section 8 con-
71 cludes.

72 **2. Institutional background: green banking policy in emerging and** 73 **transition economies**

74 The Sustainable Banking and Finance Network (SBFN)—established in
75 2012 as the Sustainable Banking Network and renamed in 2019 to reflect an
76 expanded mandate covering both banking and non-bank financial regulation—
77 is the principal vehicle through which financial-sector regulators and banking
78 associations in emerging and developing economies have coordinated the de-
79 sign of national sustainable-finance policy. Hosted and technically supported
80 by the International Finance Corporation, the network now counts roughly
81 seventy member institutions (central banks, banking regulators, and banking
82 or securities associations) drawn overwhelmingly from Sub-Saharan Africa,
83 East Asia and the Pacific, Europe and Central Asia, Latin America and
84 the Caribbean, the Middle East and North Africa, and South Asia (Inter-
85 national Finance Corporation / Sustainable Banking and Finance Network,
86 2025). Notably for our sample, the network’s membership is essentially dis-
87 joint from the European Union: not one of the sixteen EU member states in
88 our WBES sample appears on any SBFN membership roster we could locate,
89 consistent with the network’s explicit mandate to serve markets outside the
90 reach of the EU’s own sustainable-finance architecture (the Sustainable Fi-
91 nance Disclosure Regulation and the EU Taxonomy, both of which post-date
92 most of our survey waves in any case).

93 Membership itself, however, is only the network’s coarsest measure of
 94 policy adoption. SBFN’s own Multi-Tiered Progress Framework—the in-
 95 strument used in its periodic Global Progress Reports (2018, 2019, 2021,
 96 2023–2024) and country-level Progress Report addenda—scores each mem-
 97 ber jurisdiction’s actual regulatory output against several policy pillars, most
 98 centrally environmental-and-social risk management and, in later report vin-
 99 tages, ESG disclosure integration and green-finance-flow measurement, and
 100 places each jurisdiction on a progression from an initial “Commitment” or
 101 “Formulating” stage, through “Developing” or “Implementation,” to “Advanc-
 102 ing” and, for the most mature frameworks, “Consolidating.” The variation
 103 this produces across our sample is considerable and is itself part of the paper’s
 104 identifying variation, not just a robustness afterthought. Mongolia’s Bank
 105 of Mongolia was a founding SBFN member in 2012 and had already reached
 106 the network’s “Advancing” implementation stage by the time of the country’s
 107 2019 Enterprise Survey; Morocco’s Bank Al-Maghrib, a member since 2014,
 108 sat at the same stage two years later. Bangladesh Bank, also a founding 2012
 109 member, had progressed by its 2022 Enterprise Survey wave into the net-
 110 work’s ESG-integration pillar at an “Advancing” classification, while India’s
 111 Reserve Bank/Indian Banks’ Association, a later (2016) entrant, remained
 112 at a “Developing” commitment-pillar stage at its own 2022 survey. Jordan
 113 and the Kyrgyz Republic, both members since 2016 and 2018 respectively,
 114 had progressed only as far as an initial “Commitment” stage by their 2019
 115 survey waves. A further, and for identification purposes equally important,
 116 group of economies in our sample joined SBFN only after their Enterprise
 117 Survey was fielded—Kazakhstan (2021), Serbia and Ukraine (2020), Tajik-

istan (2023), and most of the Western Balkan and South Caucasus economies in our sample (2021–2024)—and are accordingly coded as non-adopters as of their survey year, even though several are now members. This ex-ante coding is deliberate: it is the policy environment a firm actually operated under at the moment its Green Economy module responses were recorded that is relevant to our test, not the policy environment that jurisdiction eventually adopted.

The economic logic behind the SBFN framework, and the reason it is a plausible channel for firm-level green investment rather than a purely symbolic commitment, rests on a straightforward supervisory mechanism: once a banking regulator formally adopts environmental-and-social risk management guidelines, commercial banks under its supervision face a compliance incentive to price environmental risk into individual lending decisions and, in many of the more advanced frameworks, to report a green-lending share to the supervisor directly. Existing evidence on China’s much-studied green credit guidelines documents exactly this channel operating at bank and firm level—reduced lending to high-polluting borrowers and improved environmental performance among firms that retain access to credit (Li et al., 2023; Huang et al., 2023; Dai et al., 2025; Gao et al., 2022)—and a smaller literature on Bangladesh’s own long-running green-banking guidelines points in the same direction (Khairunnessa et al., 2021). What neither literature has established is whether this channel is detectable across a broad, heterogeneous cross-section of SBFN-style adopters at the firm level using a common survey instrument, nor whether the strength of the channel depends on the regulatory capacity available to enforce the underlying guideline—the two

143 questions this paper turns to next.

144 **3. Literature review and hypothesis development**

145 *3.1. Access to finance and firm-level green investment*

146 A firm’s decision to install pollution-abatement equipment, retrofit ma-
147 chinery for energy efficiency, or absorb the upfront cost of an emissions-
148 monitoring system is, first and foremost, an investment decision, and in-
149 vestment decisions are shaped by whether a firm can finance them. The
150 theoretical case is close to a standard credit-constraint story: green invest-
151 ments of the kind captured in the WBES Green Economy module (heat-
152 ing and cooling upgrades, on-site renewable generation, waste-minimization
153 retrofits, energy-management systems) typically carry a large upfront capi-
154 tal outlay against a payback stream realized only over several years, which is
155 exactly the profile of investment most sensitive to a firm’s access to external
156 finance rather than to retained earnings alone. Firm-level evidence bears
157 this out directly. Ghosh and Dutta (2022), using a panel of Indian manu-
158 facturing firms, show that credit-constrained firms are measurably less likely
159 to adopt cleaner production practices even when they report an equivalent
160 willingness to comply with environmental regulation. De Haas et al. (2025)
161 document, across a large cross-country firm sample, that managerial and fi-
162 nancial barriers operate as distinct but jointly binding constraints on firms’
163 green transition investments, and Kalantzis et al. (2022) reach a closely re-
164 lated conclusion using EBRD transition-economy survey data, finding that
165 firms’ green investment is driven by a mix of financing conditions and cli-
166 mate exposure rather than by climate motivation alone—precisely the joint

167 finance-and-institutions framing this paper tests directly. Costa et al. (2024)
 168 and Truong (2026) find, respectively in OECD and Vietnamese data, that
 169 financing constraints blunt firms’ green investment response to environmen-
 170 tal policy pressure that would otherwise induce it, while Aristei and Gallo
 171 (2023) show that firms with better access to credit are both more likely to
 172 have adopted green management practices before a shock and better able to
 173 sustain them through one. B. Ullah (2025) narrows this further to unlisted
 174 SMEs specifically, showing that financial constraints bind corporate environ-
 175 mental responsibility at smaller firm scales—a firm-size channel this paper’s
 176 own causal-forest evidence (Section 5) also implicates, though, as we report
 177 there, without recovering a clean monotonic size gradient of its own. Siewers
 178 et al. (2024) extend the firm-level environmental-performance literature to
 179 a global-value-chain setting, showing that a firm’s position in international
 180 production networks shapes its environmental performance net of domes-
 181 tic financing conditions, and Bambe et al. (2026) show that environmental
 182 policy stringency itself reshapes firm efficiency in developing economies, un-
 183 derscoring that the firm-level margin, not only the aggregate or bank-level
 184 one, is where much of this literature’s true test now lies. Taken together, this
 185 literature converges on a simple first-order proposition, which any test of a
 186 green-credit-policy channel must first establish before asking whether policy
 187 moderates it:

188 **H1.** Firms with access to formal bank credit (a credit line or
 189 loan from a financial institution) are more likely to have adopted
 190 green production practices than otherwise-similar firms without
 191 such access.

192 *3.2. Does green banking policy strengthen the finance-to-green-investment*
193 *channel?*

194 H1 describes a channel, not a policy. The entire premise of the SBFN
195 framework and its national analogues—and of the much larger literature on
196 green credit guidelines in China specifically (Li et al., 2023, 2024; Huang et
197 al., 2023)—is that a supervisory mandate directing banks to price environ-
198 mental risk into lending, or to preferentially finance green activity, should
199 widen the gap in green-investment behavior between firms with and without
200 bank access relative to a counterfactual world without the policy: banks in
201 adopting jurisdictions have a compliance incentive to seek out, and price
202 favorably, exactly the borrowers undertaking the kind of investment our
203 outcome variable measures, while credit-constrained firms in the same ju-
204 risdiction see no comparable improvement in their financing terms. This
205 is a moderation hypothesis, not a simple main-effect one, and it is worth
206 being explicit that most of the existing supporting evidence—again, over-
207 whelmingly from China (Dai et al., 2025; Gao et al., 2022) plus a smaller
208 literature on Bangladesh (Khairunnessa et al., 2021), Vietnam (Nguyen and
209 Phan, 2025), and cross-country fintech-credit channels (Tran et al., 2026)—
210 is drawn from single-country studies of a single, unusually well-resourced
211 green-credit regime. Whether the same interaction holds across the far more
212 heterogeneous set of jurisdictions that have adopted an SBFN-style frame-
213 work, many of them at a far earlier and less resourced stage of implementation
214 than China’s, is untested. This motivates:

215 **H2.** The positive association between bank credit access and
216 green-practice adoption (H1) is stronger in economies that had,

217 as of the survey year, adopted an SBFN sustainable-finance policy
218 framework than in economies that had not.

219 *3.3. Institutional complementarity: does policy credibility depend on state*
220 *capacity?*

221 A green banking guideline is only as effective as the supervisory apparatus
222 behind it. This is the institutional-complementarity argument at the center of
223 the comparative-institutions, law-and-finance, and state-capacity literatures.
224 Hall and Soskice (2001)’s varieties-of-capitalism framework is the founda-
225 tional statement of the underlying logic in its most general form: formally
226 similar policy instruments produce systematically different outcomes depend-
227 ing on how they interlock with the surrounding institutional architecture—
228 a regulatory mandate that functions as intended within one configuration
229 of complementary institutions may function quite differently, or not at all,
230 transplanted into another. Applied more narrowly to financial regulation
231 specifically, the same logic runs through the law-and-finance and state-capacity
232 literatures: La Porta et al. (1998) and the subsequent finance-and-growth
233 literature (Levine, 2005; Demirgüç-Kunt and Maksimovic, 1998; Beck et al.,
234 2005, 2008) established that formally identical financial-contracting rules pro-
235 duce very different real-economy financing outcomes depending on the quality
236 of the legal and regulatory institutions that enforce them, while Acemoglu
237 et al. (2001) showed the same logic operating at the level of a country’s en-
238 tire institutional endowment. Applied to our setting, the argument is direct:
239 a bank supervisor’s environmental-risk guideline can only reshape lending
240 behavior to the extent that the supervisor has the regulatory capacity—
241 examination authority, credible sanctions, reliable disclosure enforcement—

242 to make compliance with it more than nominal. Where that capacity is
 243 weak, an SBFN framework may exist on paper (a formal “Commitment” or
 244 “Formulating” stage classification, in the network’s own terminology) with-
 245 out meaningfully altering bank lending behavior on the ground, precisely the
 246 state-capacity/policy-credibility distinction this paper is built to test. The
 247 evidence that green finance works only through capable institutions is by
 248 now reasonably direct at the country level. Shi et al. (2024), across 37 Asian
 249 economies, find that green finance improves environmental outcomes where
 250 control of corruption is strong and has little detectable effect where it is weak;
 251 Chi and Yang (2023) find China’s green-financial-system guidelines drive
 252 firms’ green transition most where state environmental governance is strong;
 253 and Jia et al. (2026) find the long-run green-credit-to-sustainable-production
 254 relationship across the BRICS economies is itself strengthened by governance
 255 quality. Each of these is an aggregate or single-jurisdiction result, and each
 256 implies a firm-level prediction—that the credit-to-green channel should re-
 257 spond to policy more sharply under stronger regulatory capacity—that has
 258 not been tested directly on firm-level data across adopting economies. That
 259 is the prediction H3 states and this paper tests. We proxy regulatory ca-
 260 pacity with the World Bank’s Worldwide Governance Indicators Regulatory
 261 Quality estimate, matched to each economy’s survey year, and hypothesize
 262 a three-way interaction:

263 **H3.** The policy-moderation effect described in H2 is itself in-
 264 creasing in regulatory quality: SBFN adoption strengthens the
 265 credit-to-green-investment channel most where the supervising
 266 jurisdiction has the institutional capacity to make the underly-

267 ing guideline credible, and least—potentially not at all—where it
268 does not.

269 Hypotheses H2 and H3 are, by construction, statements about how a
270 country-level variable reshapes a firm-level relationship. That two-level struc-
271 ture is what Section 4 argues a classical single-level specification is poorly
272 placed to adjudicate on this sample—not because the interaction terms it
273 tests are unidentified, but because a country-cluster count in the low forties
274 makes cluster-robust inference on them unreliable, and because a country-
275 fixed-effects variant would purchase within-country identification at the cost
276 of the level-2 main effects and of resting on only eight adopter economies.
277 That is what motivates the hierarchical and machine-learning estimators we
278 turn to after reporting the classical benchmark.

279 *3.4. An auxiliary hypothesis: firm-level absorptive capacity as an alternative*
280 *margin*

281 H2 and H3 are both, in the end, statements that a *country-level* char-
282 acteristic conditions the strength of the credit-to-green channel. A separate
283 strand of the financing-constraints literature suggests the conditioning vari-
284 able that matters most may instead sit at the *firm* level: B. Ullah (2025)
285 shows that financial constraints bind corporate environmental responsibility
286 specifically at smaller firm scales, consistent with a standard fixed-cost logic
287 in which the up-front capital outlay a green retrofit requires (Section 3) is
288 proportionately heavier, and therefore harder to absorb even once financed,
289 for a smaller firm than a larger one. We did not set out to test this as a fourth
290 ex-ante hypothesis on a par with H1–H3; it emerged from the causal forest’s

291 feature-importance decomposition (Section 5.4), which we report honestly
292 here as an auxiliary, exploratory hypothesis motivated by that same fixed-
293 cost logic rather than one we pre-specified:

294 **H4 (auxiliary, exploratory).** Whatever heterogeneity exists
295 in the strength of the credit-to-green-adoption channel is more
296 attributable to firm size than to either country-level institutional
297 moderator (SBFN status or regulatory quality).

298 We flag its exploratory status explicitly because it changes how its evi-
299 dence should be read: Section 5.4’s support for H4 is genuine and, we think,
300 informative, but it is evidence discovered during analysis rather than con-
301 firmed against a pre-registered prediction, and the same section reports hon-
302 estly that it does not yet extend to a clean, monotonic size gradient. H4
303 also connects this paper’s evidence to the just-transition and SME-financing
304 literature’s broader concern with who is left behind when a policy instru-
305 ment operates on an extensive banking margin rather than on the firm-scale
306 margin that determines whether a financed firm can absorb a green invest-
307 ment’s fixed costs at all—a distributional dimension of green-credit-policy
308 design this paper’s data can only gesture toward, not test directly, but which
309 we think is a natural extension for future work using firm-level data with
310 a continuous size measure and a larger, more granular firm-size distribution
311 than the Enterprise Surveys’ own three-category strata provide.

312 4. Data and methodology

313 4.1. Data sources and sample construction

314 The paper’s firm-level data come from the World Bank Enterprise Sur-
315 veys (WBES), a standardized, stratified-random-sample survey of formal-
316 sector, non-agricultural, non-extractive private firms with at least five em-
317 ployees, conducted by the World Bank’s Enterprise Analysis Unit. Our pri-
318 mary analysis sample is the World Bank’s own harmonized cross-country
319 file combining the 41 Europe and Central Asia (ECA) and Middle East and
320 North Africa (MENA) economies surveyed under a single, identical question-
321 naire between 2018 and 2020, the survey wave in which the Bank’s Green
322 Economy module—a battery of questions on emissions monitoring, energy-
323 consumption targets, and the adoption of specific climate-friendly production
324 measures—was fielded in full to every respondent. This yields 28,042 firm-
325 level observations across 41 economies (Table 2). We supplement this primary
326 sample with 15,556 additional firm observations from six further economies
327 (Bangladesh 2022, India 2022, Indonesia 2023, Peru 2023, Philippines 2023,
328 Timor-Leste 2021) in which a shortened, partially randomized version of the
329 Green Economy module was fielded; because the exact item wording and ran-
330 domized sub-module assignment differ across these six economies, we do not
331 pool them into the main harmonized outcome variable, and instead use them
332 as an out-of-region external-validity check on a narrower, honestly-matched
333 pair of indicators (CO2-emissions monitoring, available in five of the six;
334 waste-minimization adoption, available in five of the six, with different spe-
335 cific economies covered by each), reported separately in Section 5.

336 The primary sample, while broad by the standards of the existing green-

337 credit-policy literature, is not the full universe of WBES economies carrying
 338 *any* Green Economy content, and we build a second, independent sample
 339 to exploit the rest of it. The World Bank also maintains a standardized
 340 cross-country database (2006–2026) that harmonizes a common set of core
 341 variables across every Formal Sector Enterprise Survey wave it has ever
 342 fielded, 292,314 firm-level observations across 471 country-years. Within
 343 it, 160 country-years fielded between 2021 and 2026 — reaching, notably,
 344 into high-income economies entirely outside the SBFN’s remit such as Aus-
 345 tralia, Canada, Germany, and Japan — carry a harmonized minimal Green
 346 Economy item set: climate-damage exposure and, more relevantly for our
 347 outcome construct, CO2-emissions monitoring, alongside the same standard
 348 access-to-finance module (credit line/loan, overdraft, perceived finance ob-
 349 stacle) used in the primary sample. We supplement this with Bangladesh
 350 2022 and Indonesia 2023 — two economies whose Green Economy responses
 351 used an alternative randomized-module variable naming convention and so
 352 are absent from the standardized database’s harmonized columns, but which
 353 we can recover from our own extraction of the underlying country files —
 354 for a combined global sample of 162 economies and 100,115 firms (89,797
 355 with complete data on all covariates). Firm size, sector, and productivity
 356 controls for this sample are drawn from the Bank’s companion Productivity
 357 Database, merged by firm identifier. We refer to this as the *global sample*,
 358 and to CO2-emissions monitoring as its outcome variable throughout; it is
 359 the paper’s vehicle for testing whether the primary sample’s central find-
 360 ing replicates independently, at far greater scale, and under an identification
 361 strategy — country fixed effects — that the primary sample’s single-wave

362 design cannot support (Section 4).

363 Country-level data are drawn from two sources for both samples. For the
364 primary sample’s 47 economy-years, SBFN policy-adoption status, join year,
365 and progression-framework stage as of each economy’s survey year are com-
366 piled from the Sustainable Banking and Finance Network’s Global Progress
367 Reports (2018, 2019, 2021, 2023–2024), country-level Progress Report ad-
368 denda, and the network’s live country-profile data portal, cross-checked against
369 IFC press releases announcing individual member accessions; economies that
370 joined SBFN only after their WBES survey year are coded as non-adopters
371 as of that year (Section 2). For the global sample’s remaining 157 economy-
372 years, we draw membership status and join year directly from the SBFN data
373 portal’s full published member roster (67 members as of the portal’s most
374 recent update), applying the identical ex-ante rule: an economy is coded as
375 an adopter only if its join year precedes its survey year. Six economy-years
376 in the global sample (Central African Republic, Cameroon, Chad, Republic
377 of Congo, Equatorial Guinea, and Gabon, all 2023–2024) are recorded on
378 the roster under a regional entry, “Central African States,” which we inter-
379 pret as referring to the CEMAC zone’s common banking supervisor; four
380 small-island economy-years (Antigua and Barbuda, Grenada, St. Lucia, St.
381 Vincent and the Grenadines, all 2025) are similarly recorded under “East-
382 ern Caribbean States,” which we interpret as the Eastern Caribbean Central
383 Bank’s member states. Both interpretations are flagged explicitly here as
384 inferences rather than explicit per-country listings. Regulatory capacity is
385 measured with the World Bank’s Worldwide Governance Indicators (WGI)
386 Regulatory Quality estimate, matched to each economy’s exact survey year

387 (or, where not yet published, the most recent prior year available) via the
388 World Bank DataBank API; WGI does not cover Taiwan, which is accord-
389 ingly dropped from global-sample regressions.

390 A natural alternative to the cross-sectional design used throughout this
391 paper would exploit SBFN’s own staggered adoption timing—join years in
392 our sample span 2012–2024 (Section 2)—in a staggered difference-in-differences
393 or event-study design around each economy’s own adoption date, the identi-
394 fication strategy this literature most often asks for. We do not attempt this,
395 and it is worth being explicit about why rather than leaving the gap implicit.
396 The WBES Green Economy module is not a repeated panel instrument: it
397 has been fielded to a given country, in most cases, exactly once, as a special
398 one-off module attached to a single survey wave, and even the small subset of
399 economies in our sample surveyed more than once by the WBES core instru-
400 ment were not re-surveyed with a comparable Green Economy module in a
401 second wave; the six supplementary economies in our own extension sample
402 (Section 5.5) illustrate the problem directly, since even they differ from one
403 another, and from the primary sample, in exact item wording and randomized
404 sub-module assignment, precluding a consistently defined outcome measured
405 twice for the same country before and after its SBFN join date. A within-
406 country before/after comparison around each economy’s own adoption date
407 is accordingly not implementable with a consistent outcome variable in the
408 currently published WBES microdata, for this specific outcome construct.
409 This is a genuine data-availability constraint on the identification strategy
410 available to us, not a design choice; we flag it here as a first-class limitation
411 of the paper’s identification, and return to its consequences in Section 8.

412 4.2. Variable construction

413 **Outcome.** Our primary dependent variable, *green practice adoption*, is
414 a binary indicator equal to one if a firm reports having adopted at least one
415 of seven Green Economy module practices over the preceding three years:
416 more climate-friendly on-site energy generation, an energy-management sys-
417 tem, waste-minimization or recycling measures, air-pollution control mea-
418 sures, other pollution-control measures, any energy-efficiency measure, or
419 the use of on-site renewable energy. We also construct a continuous *green*
420 *adoption index*, the mean of the same seven binary items, for robustness.

421 **Treatment.** *Bank credit access* is a binary indicator equal to one if the
422 firm reports holding a line of credit or loan from a formal financial institution
423 at the time of the survey.

424 **Country-level moderators.** *SBFN member* is a binary indicator equal
425 to one if the economy had adopted a national SBFN sustainable-finance
426 policy framework as of the survey year. *Regulatory quality* is the WGI Reg-
427 ulatory Quality estimate for the survey year (continuous, standardized to
428 mean zero, unit standard deviation within the estimation sample for the
429 hierarchical and machine-learning specifications).

430 **Controls.** Firm size (small/medium/large, per the Enterprise Surveys’
431 own sampling strata), broad sector (manufacturing, services, construction),
432 log sales, an exporter dummy (any direct or indirect export share), and a
433 foreign-ownership dummy (any private foreign ownership share).

434 Table 1 lists every variable used in the analysis together with its exact
435 WBES question code and source, in the interest of full transparency about
436 variable construction given the multi-source nature of the merge.

4.3. Empirical strategy

We estimate the relationship between bank credit access and green practice adoption, and its moderation by SBFN status and regulatory quality, in five stages of increasing methodological sophistication, each addressing a specific limitation of the one before it.

Stage 1 — classical benchmark. We first estimate a pooled logistic regression,

$$\Pr(\text{Green}_{i,c} = 1) = \Lambda\left(\beta_1 \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + \delta_{s(i)}\right) \quad (1)$$

for firm i in country c , where $\mathbf{X}_{i,c}$ is the firm-control vector and $\delta_{s(i)}$ is a sector fixed effect, with standard errors clustered by country. We deliberately do not include country fixed effects in this specification, and it is worth being precise about what such a fixed effect would and would not absorb. Because SBFN_c is constant within country in a single cross-sectional wave, a country fixed effect absorbs its main effect entirely, making β_2 unidentifiable by construction. The interaction $\text{Credit}_{i,c} \times \text{SBFN}_c$, by contrast, is a firm-level regressor and remains formally identified under country fixed effects, since credit access itself varies within country; Stage 5 below relies on exactly that logic to identify the corresponding interactions on the global sample. What the primary sample cannot support is not the identification of β_3 but its precision: with country fixed effects, β_3 would be identified only by the contrast between within-country credit variation in the eight SBFN-adopter economies of this sample and that in the remaining thirty-three (Section 5), while the SBFN main-effect level shift those economies exhibit—a first-order feature of the data—would be absorbed and unreportable. We therefore estimate

460 β_2 and β_3 jointly without country fixed effects, as the classical benchmark
 461 against which Stage 2’s hierarchical alternative is compared, extended with
 462 a triple interaction for Regulatory quality_c that tests H3’s three-way claim
 463 directly.

464 **Stage 2 — Bayesian hierarchical (multilevel) model.** We re-
 465 estimate the same relationship allowing country to enter as a random rather
 466 than fixed effect, with a random slope on Credit:

$$\text{Green}_{i,c} \sim \text{Bernoulli}(p_{i,c}) \quad (2)$$

$$\begin{aligned} \text{logit}(p_{i,c}) = & (\beta_1 + u_{1,c}) \text{Credit}_{i,c} + \beta_2 \text{SBFN}_c + \beta_3 (\text{Credit}_{i,c} \times \text{SBFN}_c) \\ & + \beta_4 (\text{Credit}_{i,c} \times \text{RegQuality}_c) + \mathbf{X}'_{i,c} \boldsymbol{\gamma} + (\alpha + u_{0,c}) \end{aligned} \quad (3)$$

$$\begin{pmatrix} u_{0,c} \\ u_{1,c} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad (4)$$

467 estimated via Hamiltonian Monte Carlo (the No-U-Turn Sampler, NUTS)
 468 in PyMC, an open-source Python library for Bayesian statistical modeling
 469 (Salvatier et al., 2016). This specification is the paper’s primary vehicle for
 470 testing H2: because country enters through a variance component rather than
 471 a full set of dummies, the country-level moderators SBFN_c and RegQuality_c
 472 retain identified main effects, while the random slope $u_{1,c}$ still absorbs unob-
 473 served country heterogeneity in the credit-green relationship and, together
 474 with partial pooling across countries, gives more defensible inference than
 475 cluster-robust standard errors computed over only 41 clusters (Cameron and
 476 Miller, 2015). It carries two cross-level interactions: β_3 tests H2 directly, and
 477 β_4 tests whether regulatory quality on its own moderates the credit-green
 478 relationship. We are explicit that β_4 is not a test of H3. H3 asks whether

the SBFN moderation in H2 is itself conditional on regulatory quality, which is a three-way claim requiring a $\text{Credit}_{i,c} \times \text{SBFN}_c \times \text{RegQuality}_c$ term; this specification does not include one, and we do not add it here because the primary sample’s eight adopter economies cannot support a credibly estimated three-way interaction under partial pooling. H3 is instead tested directly by Stage 1’s fully saturated triple interaction above (Table 3, column M3) and, on an independent sample, by Stage 5’s no-fixed-effects specification (Section 5.7), both of which retain that term.

Stage 3 — causal forest / double machine learning. To avoid imposing a linear interaction functional form on how the treatment effect of credit access varies across the institutional landscape, we additionally estimate a Causal Forest with Double Machine Learning (Wager and Athey, 2018; Chernozhukov et al., 2018; Athey et al., 2019), using gradient-boosted nuisance models for the outcome and treatment propensity, and firm size, export status, SBFN status, and regulatory quality as effect-modifying features. This lets us recover conditional average treatment effects (CATEs)—the estimated effect of credit access on green adoption, separately for SBFN members versus non-members, and separately by regulatory-quality tercile—without imposing that the moderation take the single-coefficient interaction form assumed in Stages 1–2. We use “causal” here in the qualified sense standard in this literature: the forest’s identifying assumption is unconfoundedness conditional on the observed covariate set (firm size, export status, SBFN status, regulatory quality), not an exogenous source of variation in credit access itself. This is exactly the same assumption underlying the logit and hierarchical specifications; Stage 3 relaxes the functional form imposed on

504 that assumption from linear to fully flexible, but it does not relax the as-
505 sumption, and the CATEs reported below should be read accordingly as
506 doubly-robust conditional associations rather than causal effects identified
507 off a natural experiment.

508 **Stage 4 — small-sample supplementary check.** We additionally
509 re-estimate a reduced-form version of the credit-green relationship using a
510 waste-minimization-adoption outcome, available in four economies (India
511 2022, Indonesia 2023, Peru 2023, Timor-Leste 2021) whose Green Economy
512 modules asked this item under directly comparable wording. This outcome
513 is not part of the standardized cross-country database underlying Stage 5
514 below, so it offers genuinely independent, if small-sample, supplementary
515 evidence.

516 **Stage 5 — global-sample replication with country fixed effects.**
517 Finally, and centrally, we re-estimate the full specification on the 162-economy
518 global sample described in Section 4, using CO2-emissions monitoring as the
519 outcome. Because this sample spans far more country clusters than the
520 primary sample, we can now afford a country-fixed-effects specification: δ_c
521 absorbs SBFN_c and RegQuality_c 's main effects (collinear with the fixed effect
522 by construction, since both are constant within country-year) while leaving
523 their firm-level interactions with $\text{Credit}_{i,c}$ fully identified, since credit access
524 itself varies within country. This is a materially sharper test of H2 than Stage
525 1's clustered-SE approach, on an entirely independent sample and outcome
526 from Stages 1–3. The fixed-effects specification carries the two two-way in-
527 teractions only, so we report it alongside a no-fixed-effects specification for
528 two reasons: direct comparability with the primary sample's Table 3, and

529 because that specification, like Table 3’s M3, retains the triple interaction
530 H3 requires.

Table 1: Variable definitions and sources.

Variable	Definition	WBES code	Source
green_adoption_binary	=1 if firm adopted ≥ 1 of 7 Green Economy practices in last 3 years	BMGc23b, c23d, c23e, c23f, c23j, c25, e5	WBES Green Economy Module (2018–2020)
fin_has_credit_line	=1 if firm has a line of credit or loan	K8	WBES Access to Finance Module
sbfn_member	=1 if SBFN policy adopted as of survey year	n/a	SBFN Global Progress Reports; data.sbfnetwork.org
wgi_regulatory_quality	WGI Regulatory Quality estimate, survey year	n/a	World Bank DataBank API (GOV_WGI_RQ_EST)
size_cat	Firm size stratum (Small/Medium/Large)	A6A	WBES Sampling Frame
sector_broad	Broad sector (Manufacturing/Services/Construction)	A4AX	WBES Sampling Frame
log_sales	Log of total annual sales	D2	WBES General Information Module
foreign_owned_dummy	=1 if any foreign ownership share $> 0\%$	B2B	WBES Ownership Module
exporter_dummy	=1 if any export sales share $> 0\%$	D3B, D3C	WBES Sales Module

531 5. Results

532 5.1. Descriptive patterns

Table 2: Sample composition, by economy.

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Albania 2019	377	2019	0	0.09	0.609	0.407
Armenia 2020	546	2020	0	0.21	0.684	0.482
Azerbaijan 2019	225	2019	0	-0.10	0.418	0.180
Belarus 2018	600	2018	0	-0.79	0.662	0.425
Bosnia and Herze- govina 2019	362	2019	0	-0.38	0.581	0.565
Bulgaria 2019	772	2019	0	0.46	0.604	0.467
Croatia 2019	404	2019	0	0.38	0.708	0.448
Cyprus 2019	240	2019	0	1.04	0.746	0.711
Czechia 2019	502	2019	0	1.12	0.725	0.526
Egypt 2020	3,075	2020	1	-0.64	0.303	0.065
Estonia 2019	360	2019	0	1.54	0.738	0.412
Georgia 2019	581	2019	1	0.88	0.441	0.501
Greece 2018	600	2018	0	0.29	0.825	0.499
Hungary 2019	805	2019	0	0.46	0.673	0.537
Italy 2019	760	2019	0	0.70	0.506	0.230
Jordan 2019	601	2019	1	0.16	0.471	0.248
Kazakhstan 2019	1,446	2019	0	0.11	0.519	0.257
Kosovo 2019	271	2019	0	-0.38	0.698	0.425
Kyrgyz Republic 2019	360	2019	1	-0.42	0.606	0.299
Latvia 2019	359	2019	0	1.00	0.824	0.386

Economy	N firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
Lebanon 2019	532	2019	0	-0.56	0.452	0.365
Lithuania 2019	358	2019	0	0.98	0.678	0.342
Malta 2019	242	2019	0	0.83	0.901	0.479
Moldova 2019	360	2019	0	-0.11	0.619	0.328
Mongolia 2019	360	2019	1	-0.07	0.625	0.492
Montenegro 2019	150	2019	0	0.29	0.387	0.576
Morocco 2019	1,096	2019	1	0.03	0.478	0.251
North Macedonia 2019	360	2019	0	0.19	0.572	0.508
Poland 2019	1,369	2019	0	0.83	0.569	0.403
Portugal 2019	1,062	2019	0	1.00	0.812	0.600
Romania 2019	814	2019	0	0.36	0.438	0.474
Russia 2019	1,323	2019	0	-0.35	0.503	0.310
Serbia 2019	361	2019	0	0.06	0.654	0.585
Slovak Republic 2019	429	2019	0	0.73	0.782	0.386
Slovenia 2019	409	2019	0	0.88	0.793	0.674
Tajikistan 2019	352	2019	0	-0.80	0.523	0.194
Tunisia 2020	615	2020	1	-0.11	0.391	0.413
Türkiye 2019	1,663	2019	1	-0.02	0.346	0.345
Ukraine 2019	1,337	2019	0	-0.29	0.715	0.244
Uzbekistan 2019	1,239	2019	0	-0.92	0.751	0.240

Economy	<i>N</i> firms	Year	SBFN	Reg. qual.	Green rate	Credit rate
West Bank and Gaza 2019	365	2019	0	-0.15	0.428	0.178

Note: Reg. qual. is the WGI Regulatory Quality estimate for the survey year. Green rate is the unconditional share of firms reporting adoption of at least one Green Economy practice. Credit rate is the share of firms reporting a bank credit line or loan.

533 Table 2 reports the full 41-economy sample composition; firm counts range
534 from 150 (Montenegro) to 3,075 (Egypt), and the raw, unconditional green-
535 practice-adoption rate ranges more than three-fold across economies, from
536 30.3% (Egypt) to 90.1% (Malta). Figure 1 maps this variation alongside
537 SBFN policy status: 8 of the 41 primary-sample economies had adopted
538 an SBFN framework as of their survey year (Georgia, Egypt, Jordan, Kyr-
539 gyz Republic, Mongolia, Morocco, Tunisia, and Türkiye)—13 of the full 47-
540 economy sample once the five (of six) SBFN-member small-sample supple-
541 mentary economies are counted—and every European Union member state in
542 the sample is, consistent with SBFN’s mandate, a non-adopter. At the firm
543 level, bank credit access correlates positively with the green-adoption index
544 ($r = 0.18$) but is essentially uncorrelated with a firm’s own perceived finance
545 obstacle ($r = -0.02$)—a first hint, well before any regression, that realized
546 access to credit rather than perceived financing difficulty is what tracks green
547 behavior, and the two are evidently not interchangeable measures of the same
548 underlying constraint.

549 Figure 2 previews the paper’s central finding graphically before a sin-
550 gle regression is estimated: plotting each economy’s green-adoption rate

551 against its regulatory-quality score and fitting separate lines for SBFN mem-
 552 bers and non-members produces two lines of very similar, modestly positive
 553 slope, separated by a roughly 15–20 percentage-point vertical gap—SBFN-
 554 member economies show systematically *lower* raw green-adoption rates at
 555 every level of regulatory quality, not a steeper relationship between regu-
 556 latory quality and adoption. That combination—a level difference with no
 557 slope difference—is exactly the signature the regression evidence below con-
 558 firms formally.

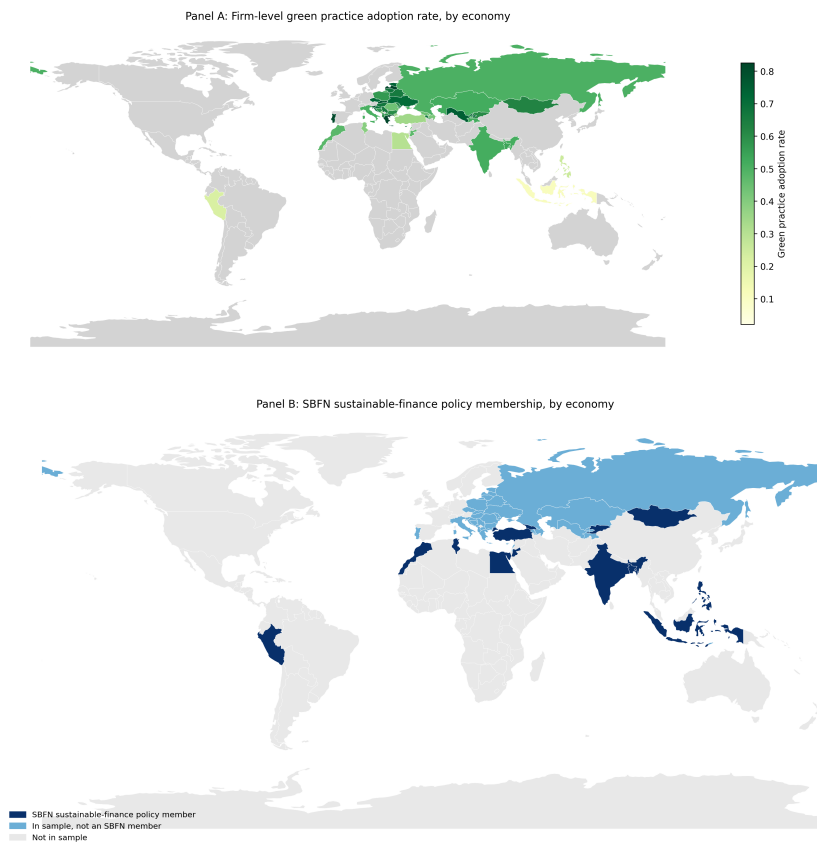


Figure 1: Green practice adoption and SBFN policy membership, by economy.

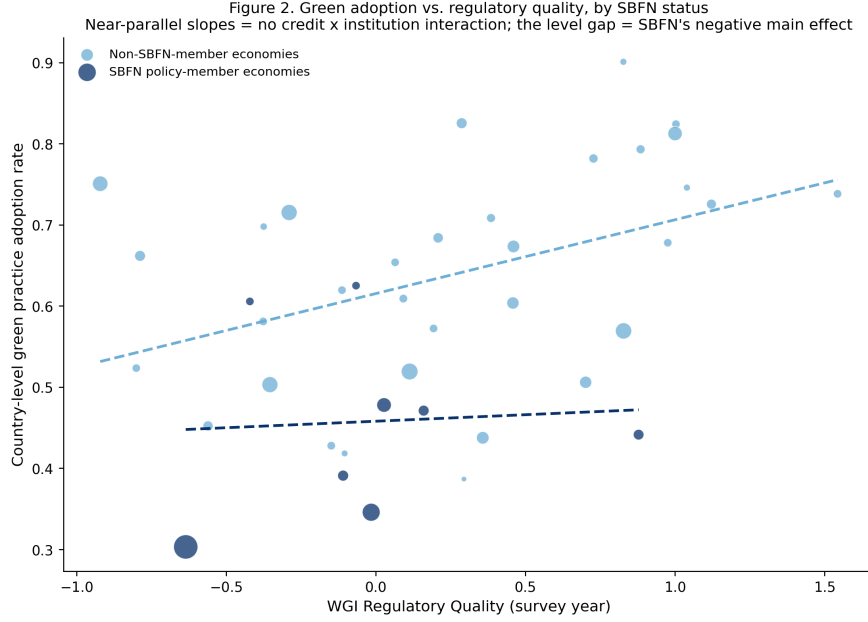


Figure 2: Green adoption vs. regulatory quality, by SBFN status.

5.2. Baseline classical benchmark

Table 3 reports the pooled logistic baseline. Column M1 confirms H1 on its own terms: firms with a bank credit line are significantly more likely to have adopted a green practice ($b = 0.571$, $p < 0.001$; average marginal effect = 12.95 percentage points, 95% CI [7.7, 18.2])—holding a formal credit line is associated with roughly the same increase in the probability of green adoption as moving from a small to a large firm, and a somewhat larger increase than being an exporter. Column M2 adds SBFN status and its interaction with credit access: the SBFN main effect is large, negative, and precisely estimated ($b = -0.991$, $p < 0.001$), while the credit $\times$ SBFN interaction is small and statistically indistinguishable from zero ($b = 0.049$, $p = 0.773$). Column M3 adds the fully saturated triple interaction with regulatory quality, and

571 it is this column, not the hierarchical model of Section 5.3, that carries the
572 paper’s direct test of H3: the credit $\times$ SBFN $\times$ regulatory-quality term is
573 small, wrong-signed relative to H3’s prediction, and nowhere near significance
574 ($b = -0.104$, $se = 0.272$, $p = 0.702$). The rest of M3 tells the same story
575 at greater length: every interaction term involving credit access is small
576 and insignificant, while regulatory quality’s uninteracted relationship with
577 adoption is positive, if only marginally significant in this particular fully-
578 saturated specification ($b = 0.278$, $p = 0.110$).

579 As set out in Section 4, adding country fixed effects here would absorb the
580 SBFN and regulatory-quality main effects entirely, while leaving the interac-
581 tion terms identified but reliant on the contrast between within-country credit
582 variation in this sample’s eight adopter economies and its thirty-three non-
583 adopters. Table 3 accordingly reports main effects and interactions jointly,
584 without country fixed effects; what it shows is a genuine test of H2 and, in
585 M3, of H3, not a placeholder—and that test returns no interaction on ei-
586 ther. The question the remaining two stages are built to answer is whether
587 that null result reflects a real absence of policy moderation, or an artifact
588 of forcing a two-level question through a single-level model with country-
589 clustered standard errors computed over only 41 clusters, a setting in which
590 such standard errors are known to be unreliable (Cameron and Miller, 2015).

591 *5.3. Bayesian hierarchical model — primary test of H2*

592 The hierarchical specification—country-varying random slopes on credit
593 access, cross-level interactions with SBFN status and standardized regulatory
594 quality, four chains of 1,000 post-warmup draws each, all $\hat{R} \leq 1.01$ and
595 ess_{bulk} between 797 and 2,270—is built specifically to give H2 a fair hearing

Table 3: Baseline classical logit regressions (dependent variable: green practice adoption, binary). Country-clustered standard errors in parentheses.

	M1	M2	M3
Credit access	0.571*** (0.123)	0.459*** (0.090)	0.373*** (0.085)
SBFN member		-0.991*** (0.178)	-0.844*** (0.209)
Credit $\times$ SBFN		0.049 (0.168)	0.019 (0.140)
Regulatory quality			0.278 (0.174)
Credit $\times$ Reg. quality			0.144 (0.142)
SBFN $\times$ Reg. quality			0.053 (0.261)
Credit $\times$ SBFN $\times$ Reg. quality			-0.104 (0.272)
Log sales	0.061 (0.039)	0.041 (0.034)	0.068** (0.032)
Foreign owned	0.219 (0.143)	0.244* (0.139)	0.220 (0.143)
Exporter	0.537*** (0.092)	0.471*** (0.074)	0.412*** (0.079)
Sector, size controls	Yes	Yes	Yes
N	23,914	23,914	23,914
Pseudo R^2	0.058	0.089	0.093

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

under inference the 41-cluster classical specification cannot deliver, and to test regulatory quality’s own two-way moderation alongside it. H3’s three-way claim is not tested here (Section 4); it is tested in Table 3’s M3 above and in Section 5.7 below, and returns a null in both. This specification does not change the conclusion either. Table 4 reports the full posterior summary; three results stand out.

First, the firm-level credit effect (H1) survives unchanged in substance: posterior mean 0.33 (89% equal-tailed interval [0.18, 0.49]), comfortably excluding zero. Second, both country-level institutional variables retain clean, precisely estimated *main* effects on the baseline adoption rate—SBFN membership net-negative (-0.57, 89% ETI [-1.00, -0.09]), regulatory quality net-positive (0.34, 89% ETI [0.16, 0.53])—reproducing, with proper hierarchical uncertainty quantification rather than an absorbed or clustered approximation, exactly the level-shift pattern Figure 2 shows visually. Third, and centrally for H2, neither cross-level interaction is distinguishable from zero: credit $\times$ SBFN, 0.072 (89% ETI [-0.26, 0.40]); credit $\times$ regulatory quality (standardized), 0.004 (89% ETI [-0.13, 0.14])—a near-exact zero on the latter, not merely an imprecise one. The first of these is the direct H2 test; the second establishes that regulatory quality does not moderate the credit-green channel on its own, which is a necessary condition for the mechanism H3 posits without being a test of H3 itself.

We read this as an honest and, we think, informative finding rather than a disappointing one: giving the moderation hypothesis a hierarchical model built for precisely this two-level question still returns no interaction, while sharpening (relative to the pooled baseline) the confidence with which we

621 can state that SBFN status and regulatory quality shift the *level* of green
622 adoption directly.

Table 4: Bayesian hierarchical model, posterior summary (four chains, 1,000 post-warmup draws each).

Parameter	Mean	SD	89% ETI low	89% ETI high	$\hat{R}$
Intercept	-1.93	0.246	-2.30	-1.50	1.00
Credit access	0.33	0.097	0.18	0.49	1.00
SBFN member	-0.57	0.300	-1.00	-0.09	1.01
Credit $\times$ SBFN	0.072	0.211	-0.26	0.40	1.00
Reg. quality (std.)	0.344	0.116	0.16	0.53	1.00
Credit $\times$ Reg. quality	0.004	0.087	-0.13	0.14	1.00

623 5.4. Causal forest / heterogeneous treatment effects

624 Table 5 and Figure 5 report the Causal Forest DML estimates, which im-
625 pose no linear-interaction functional form and let the treatment effect vary
626 flexibly across firm and country covariates. The overall average treatment
627 effect of credit access is 0.073 (95% CI [-0.040, 0.186])—directionally consis-
628 tent with, though smaller in magnitude and less precisely estimated than,
629 the logit average marginal effect of 0.130 in Section 5.2, reflecting the more
630 conservative variance properties of a doubly-robust, cross-fitted nonparamet-
631 ric estimator relative to a parametric logit on the same data. Critically,
632 the conditional average treatment effects triangulate with Section 5.3 rather
633 than contradicting it: the estimated effect is 0.072 for firms in non-SBFN
634 economies versus 0.076 for firms in SBFN economies, and 0.075 / 0.070 /

0.075 across the low/middle/high regulatory-quality terciles respectively—
three different institutional cuts, three essentially flat effects, all visibly overlapping in Figure 5. Two entirely independent estimators, one parametric and hierarchical, one nonparametric and forest-based, agree on the absence of institutional moderation.

The forest’s feature-importance decomposition adds a genuinely new result rather than only confirming the null, and speaks directly to the auxiliary H4 introduced in Section 3.4: of the four candidate effect-modifiers, firm size (log sales) accounts for 75.6% of the model’s explained heterogeneity, versus 19.9% for regulatory quality, 2.6% for SBFN status, and 1.9% for exporter status. Whatever heterogeneity exists in how strongly credit access predicts green adoption is, on this evidence, primarily a firm-level rather than a country-institutional phenomenon, supporting H4. Collapsing this into the Enterprise Surveys’ own coarse size strata, however, does not reveal a clean monotonic gradient: the estimated effect is 0.080 for small firms, 0.075 for medium firms, and 0.057 for large firms (Table 6; all with wide, overlapping confidence intervals spanning zero), which if anything runs opposite to a simple “larger firms better absorb a green investment’s fixed costs” story rather than confirming it. Read alongside the 75.6% importance share for the continuous size measure, we take this as evidence that the heterogeneity the forest is detecting is more granular or non-monotonic across the firm-size distribution than a three-category collapse can reveal, rather than evidence for any particular direction of a size gradient—a qualification we report explicitly rather than paper over with a tidier-sounding but unsupported claim. This redirects, rather than closes, the paper’s institutional-moderation question,

660 and we return to it in Section 7.

Table 5: Causal forest DML: average treatment effect, CATEs, and feature importances.

Estimate	Value	95% CI low	95% CI high
Overall ATE	0.0733	-0.0396	0.1862
CATE SBFN non-member	0.0723	-0.0300	0.1746
CATE SBFN member	0.0758	-0.0591	0.2107
CATE reg. quality tercile: low	0.0753	-0.0568	0.2075
CATE reg. quality tercile: mid	0.0697	-0.0355	0.1749
CATE reg. quality tercile: high	0.0749	-0.0203	0.1701
Feature	Importance		
SBFN member	0.0257		
Regulatory quality	0.1993		
Log sales	0.7562		
Exporter dummy	0.0189		

661 *5.5. Small-sample supplementary check: waste-minimization adoption*

662 Before turning to the global-sample replication that is this paper’s second
663 major test (Section 5.7), we report a smaller supplementary check using
664 waste-minimization adoption—an outcome not captured in the standardized
665 cross-country database underlying the global sample, and so a genuinely
666 independent (if small-sample) data point—available in four economies whose
667 Green Economy modules asked this item under directly comparable wording:
668 India 2022, Indonesia 2023, Peru 2023, and Timor-Leste 2021 (descriptive
669 rates in Supplementary Table S1). Three of these four are themselves SBFN

Table 6: Causal forest CATEs by firm-size category.

Size category	CATE	95% CI low	95% CI high
Small	0.0797	-0.0409	0.2004
Medium	0.0747	-0.0344	0.1838
Large	0.0569	-0.0428	0.1565

members, so this sample cannot re-test the SBFN-adoption contrast; what it can test is whether the credit-to-green channel documented in Sections 5.2–5.4 generalizes beyond the ECA-MENA sample. It does: pooled across the four economies, having an overdraft facility is associated with significantly higher odds of waste-minimization adoption ($b = 0.422$, $p < 0.001$, $N = 9,038$, 4 countries)—a preview, on a different continent and half-decade, of the much larger replication in Section 5.7.

5.6. Additional robustness

Supplementary Table S2 subjects the primary-sample finding to four further checks (sector fixed effects, firm controls, country-clustered standard errors; no country fixed effects, for the reason given in Section 4). Replacing the binary outcome with the continuous seven-item green-adoption index reproduces the same pattern by OLS. Replacing bank credit access with two alternative finance measures produces one point of nuance: using overdraft-facility access, the overdraft $\times$ SBFN interaction is negative and significant ($b = -0.346$, $p = 0.031$)—the one interaction term in the paper that clears a conventional significance threshold, and it runs opposite to H2’s predicted direction—whereas using the reverse-coded perceived finance-obstacle mea-

688 sure, both the main effect and its SBFN interaction are precisely zero. Split-
 689 ting the sample into manufacturing and services subsamples shows the credit-
 690 access main effect and the null credit $\times$ SBFN interaction replicate within
 691 each sector separately. We read the overdraft result as suggestive rather than
 692 conclusive: it is one significant coefficient among more than a dozen interac-
 693 tion tests across the paper, and a Benjamini–Hochberg false-discovery-rate
 694 correction applied to that full family (Supplementary Table S3) pushes its
 695 raw $p = 0.031$ to $p = 0.403$ —no coefficient in the family survives at a false
 696 discovery rate of 0.10 or even 0.40, exactly what a pure false-discovery process
 697 would be expected to produce roughly one time in twenty.

698 *5.7. Global-sample replication with country fixed effects*

699 The preceding five subsections establish the paper’s central finding on the
 700 primary sample. We now subject that finding to what we consider the most
 701 demanding test available to us: complete re-estimation on an independently
 702 assembled, 162-economy global sample (Section 4), using a different outcome
 703 variable (CO2-emissions monitoring rather than the seven-item composite),
 704 different survey years (2021–2026 rather than 2018–2020), and, for the first
 705 time in the paper, a country-fixed-effects specification that this much larger
 706 set of clusters can support. Figure 3 maps the combined reach of the two
 707 samples together: 166 distinct economies, spanning every region and, through
 708 the global sample specifically, extending for the first time in this literature
 709 into high-income economies with no SBFN involvement at all.

710 Table 7 reports two specifications. Column M1, without country fixed ef-
 711 fects, is the direct global-sample analogue of Table 3’s primary-sample base-
 712 line: credit access remains positive and significant ($b = 0.198$, $p = 0.039$,

⁷¹³ $N = 89,797$, 159 countries), while the SBFN main effect ($b = -0.171$,
⁷¹⁴ $p = 0.353$), the credit $\times$ SBFN interaction ($b = 0.278$, $p = 0.430$), and
⁷¹⁵ every regulatory-quality interaction are statistically indistinguishable from
⁷¹⁶ zero. Like Table 3's M3, this specification is fully saturated and so retains
⁷¹⁷ the credit $\times$ SBFN $\times$ regulatory-quality term H3 requires, which is likewise
⁷¹⁸ insignificant; Table 7 tabulates the focal two-way terms only. H3 therefore re-
⁷¹⁹ ceives a direct, fully saturated test on both firm-level samples, and returns a
⁷²⁰ null on each. Column M2 imposes country fixed effects, absorbing the SBFN
⁷²¹ and regulatory-quality main effects by construction (they are collinear with
⁷²² δ_c , since both are constant within a given country-year) while leaving their
⁷²³ firm-level interactions with credit access fully identified, because credit access
⁷²⁴ itself varies within country. This is a sharper test than anything the primary
⁷²⁵ sample's single 2018–2020 wave can support, and it returns the same answer
⁷²⁶ with even more precision: credit access is strongly significant ($b = 0.249$,
⁷²⁷ $p < 0.001$), while the credit $\times$ SBFN interaction ($b = 0.506$, $p = 0.205$) and
⁷²⁸ the credit $\times$ regulatory-quality interaction ($b = -0.044$, $p = 0.346$) remain
⁷²⁹ null.

⁷³⁰ We regard this as the paper's most important robustness result. It is not
⁷³¹ a re-analysis of the same data under a different label: it is a different outcome
⁷³² construct, a different set of economies (only a handful of which overlap with
⁷³³ the primary sample), a different half-decade, and an econometric specification
⁷³⁴ — country fixed effects — that is only available at all because this second
⁷³⁵ sample has enough clusters to support it. That the qualitative conclusion
⁷³⁶ is identical across both samples and across the full set of four estimators
⁷³⁷ reported in this paper (classical logit, Bayesian hierarchical, causal forest, and

738 now country-fixed-effects logit) is, we think, considerably stronger evidence
 739 for the paper’s central claim than either sample could offer alone.

Figure 3. Combined sample coverage: 166 economies (41 primary rich-module + 162 global minimal-indicator, net of overlap)

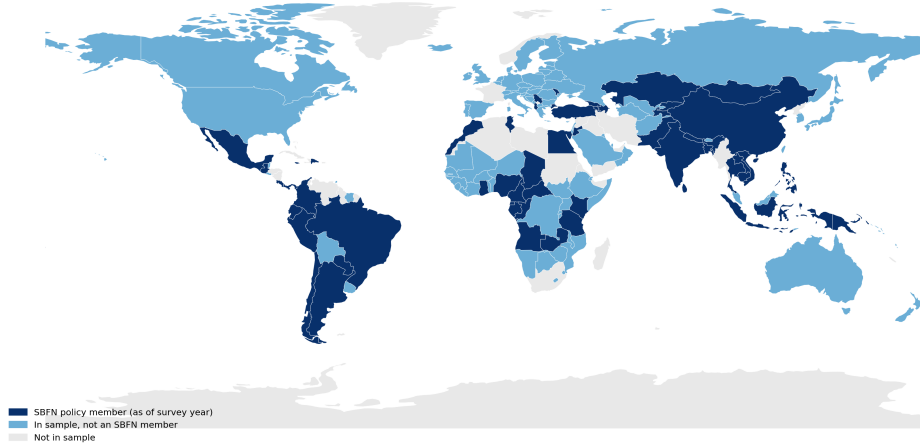


Figure 3: Combined sample coverage: 166 economies (41 primary + 162 global, net of overlap), shaded by SBFN status.

740 5.8. Results summary across all specifications

741 Sections 5.2–5.7 report ten tables spanning four estimators and two inde-
 742 pendent samples. Figure 4 consolidates every coefficient bearing on H1–H3
 743 into a single view. Panel A shows the credit-access effect (H1): every point
 744 estimate is positive and every confidence interval excludes zero, in both sam-
 745 ples and all four estimators. Panels B and C show the credit $\times$ SBFN and
 746 credit $\times$ regulatory-quality interactions (H2, H3): every single confidence in-
 747 terval, across both samples and every estimator, straddles zero. Figure 5
 748 shows the same non-result from a completely different angle—the causal
 749 forest’s conditional average treatment effects broken out by SBFN status,
 750 regulatory-quality tercile, and firm-size category all overlap both each other

Table 7: Global-sample regressions (dependent variable: CO2-emissions monitoring, binary). Country-clustered standard errors in parentheses; M2's SBFN and regulatory-quality main effects are absorbed by country fixed effects and are not separately estimable.

	M1: no country FE	M2: country FE
Credit access	0.198** (0.096)	0.249*** (0.057)
SBFN member	-0.171 (0.184)	— (absorbed)
Credit $\times$ SBFN	0.278 (0.352)	0.506 (0.399)
Regulatory quality (std.)	0.155 (0.096)	— (absorbed)
Credit $\times$ Reg. quality	0.092 (0.089)	-0.044 (0.047)
Log sales	0.127*** (0.027)	0.288*** (0.028)
Foreign owned	0.621*** (0.083)	0.528*** (0.046)
Exporter	0.456*** (0.094)	0.335*** (0.044)
Country fixed effects	No	Yes
N	89,797	89,797
Countries	159	159

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Taiwan dropped (no WGI coverage).

and the overall average treatment effect. No subgroup split, by any variable examined in this paper, recovers a distinguishable effect.

Figure 4. Coefficient estimates across every specification and both samples
(Panel A: robust and significant everywhere; Panels B-C: null everywhere)

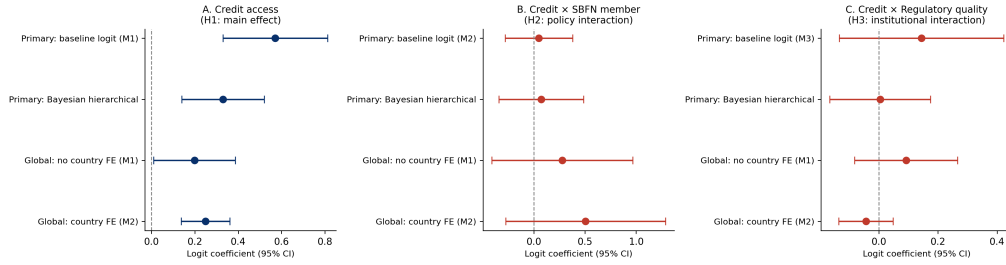


Figure 4: Coefficient estimates across every specification and both samples.

6. Supplementary evidence: a country-level staggered event-study on macro environmental outcomes

Sections 5.2–5.8 establish this paper’s central finding using firm-level cross-sectional data, for the data-availability reasons given in Section 4: the WBES Green Economy module cannot support a staggered difference-in-differences or event-study design around each economy’s own SBFN adoption date, because it has not been fielded as a repeated panel instrument. That constraint applies to the firm-level finance-to-green-investment channel specifically; it does not apply to a different, complementary question that the same SBFN adoption-timing variation can answer directly: does a country’s aggregate environmental trajectory shift around its own SBFN adoption date? This section builds and reports that test, using an entirely independent data source and unit of analysis, precisely because it is the identification

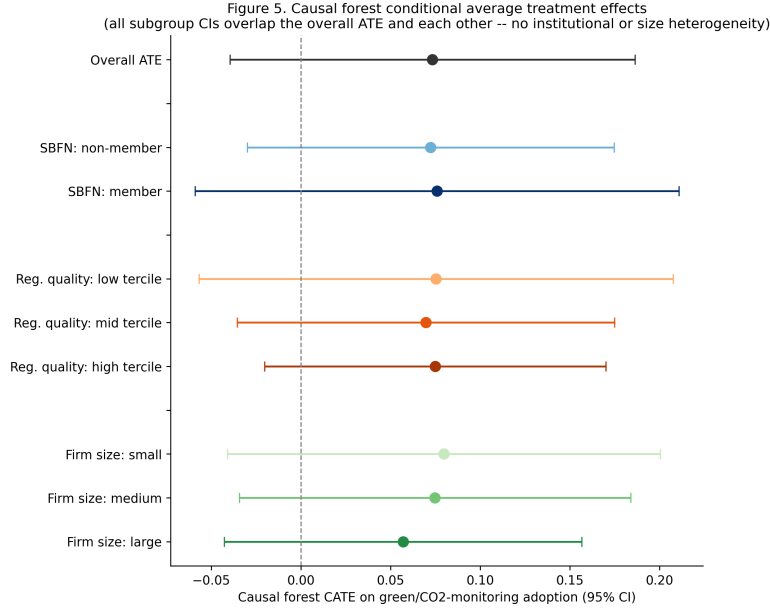


Figure 5: Causal forest conditional average treatment effects, by subgroup.

strategy this literature most often asks for, and because a genuine answer,
however it comes out, is more useful than leaving the question unaddressed.

6.1. Data and design

We assemble a country-year panel spanning every World Bank member
economy with a non-aggregate ISO3 code (217 economies), 2000–2024. Treat-
ment timing is each economy’s SBFN join year, drawn from the same SBFN
data portal membership roster used to construct the global sample (Sec-
tion 4) and cross-checked against the global-sample SBFN coding used in
Section 5.7 (zero disagreements across 61 overlapping economies); economies
that have never joined SBFN, as of the roster’s most recent update, serve as
the never-treated comparison group. This yields 68 ever-treated economies
(join years 2012–2024, the same real staggered-adoption variation described

778 in Section 2) and 149 never-treated economies. Outcome data come from
 779 the World Bank’s World Development Indicators, retrieved via the World
 780 Bank DataBank API: renewable energy consumption (% of total final en-
 781 ergy consumption, `EG.FEC.RNEW.ZS`) and CO2 emissions per capita (AR5
 782 GWP-consistent series, `EN.GHG.CO2.PC.CE.AR5`), both available for the large
 783 majority of economies from 2000 through at least 2021–2024. We addition-
 784 ally draw on GDP per capita (constant 2015 US\$, `NY.GDP.PCAP.KD`) as a
 785 covariate, to net out the obvious confound that SBFN-adopting economies
 786 are disproportionately still-industrializing, faster-growing economies whose
 787 emissions profiles are shaped by their development stage independently of
 788 any banking-sector policy.

789 We estimate the effect of SBFN adoption on each outcome using the
 790 Callaway and Sant’Anna (2021) staggered-adoption difference-in-differences
 791 estimator (Callaway and Sant’Anna, 2021), which recovers group-time av-
 792 erage treatment effects for each adoption cohort separately and aggregates
 793 them into an overall average treatment effect on the treated (ATT) and
 794 an event-study profile. We report this estimator, rather than a standard
 795 two-way-fixed-effects (TWFE) regression, because TWFE is now well known
 796 to produce badly biased estimates under staggered treatment timing when
 797 already-treated units are implicitly used as controls for later-treated units
 798 (Goodman-Bacon, 2021); we report a naive TWFE specification alongside
 799 the Callaway-Sant’Anna estimates specifically to illustrate the size of that
 800 bias in our own data, rather than as a candidate preferred estimate. For
 801 each outcome we report both an unconditional specification and a doubly-
 802 robust specification conditioning on log GDP per capita, and we check the

doubly-robust specification's sensitivity to the choice of comparison group, using never-treated economies as our preferred comparison group throughout and not-yet-treated economies (those that will eventually adopt but had not yet, as of the relevant relative period) as a robustness check.

6.2. Results

Table 8 reports the overall ATT for both outcomes across four specifications, including a comparison-group robustness check; Figure 6 plots the full event-study profile (doubly-robust, GDP-per-capita-controlled, never-treated specification) for both outcomes.

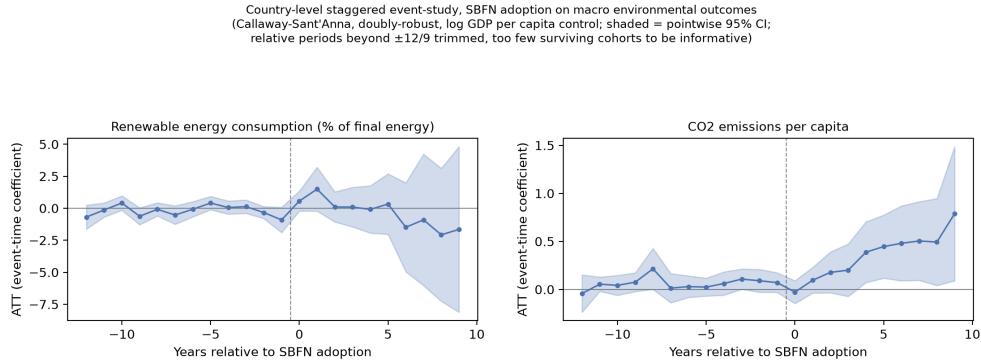


Figure 6: Country-level staggered event-study, doubly-robust specification with log GDP per capita control. Shaded bands are pointwise 95% confidence intervals; relative periods beyond $\pm 12/9$ are trimmed for having too few surviving cohorts to be informative.

Two results, read together, do not support treating SBFN adoption as a country-level lever that visibly improves aggregate environmental outcomes – which is itself informative, since this is the identification strategy best suited to answer that specific question directly. First, renewable energy share shows no effect distinguishable from zero in either specification, and the doubly-

Table 8: Country-level staggered event-study: overall average treatment effect of SBFN adoption on macro environmental outcomes. Callaway-Sant’Anna estimator, 999 bootstrap iterations; naive TWFE reported for comparison only. Full country-year panel, 2000–2024.

Outcome	Specification	ATT	SE	95% CI
Renewable energy %	Callaway-Sant’Anna, no covariates, never-treated	-1.192	0.802	[-2.763, 0.380]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.674	0.916	[-1.122, 2.470]
Renewable energy %	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.653	0.919	[-1.147, 2.454]
Renewable energy %	Naive TWFE (post dummy, country+year FE)	-4.672***	1.127	[-6.882, -2.463]
CO2 per capita	Callaway-Sant’Anna, no covariates, never-treated	0.631***	0.179	[0.280, 0.982]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., never-treated	0.349**	0.156	[0.044, 0.655]
CO2 per capita	Callaway-Sant’Anna, doubly-robust, log GDP p.c., not-yet-treated	0.313**	0.148	[0.023, 0.603]
CO2 per capita	Naive TWFE (post dummy, country+year FE)	0.901***	0.204	[0.500, 1.302]

*** 95% CI excludes zero; ** 95% CI excludes zero, narrower margin. Underlying data and code in the replication package.

817 robust event-study profile (Figure 6, left panel) is flat and precisely estimated
818 on both sides of the adoption date: a clean null, not an underpowered one.

819 Second, CO2 emissions per capita shows a positive (i.e., the “wrong-
820 signed” direction if SBFN adoption reduced emissions) and statistically sig-
821 nificant association with adoption in every specification, but three features
822 of the result argue against reading it causally. It shrinks by roughly 45%
823 (0.631 to 0.349) once we condition on log GDP per capita, indicating that
824 a substantial share of the raw association is attributable to exactly the
825 development-stage confound we set out to net out. The doubly-robust event-
826 study profile (Figure 6, right panel) shows a small, mostly flat pre-period
827 with only two marginally significant leads, followed by a post-period effect
828 that grows smoothly and monotonically from near zero at adoption to its
829 largest magnitude nine to twelve years out – a continuing-divergence pattern
830 more consistent with SBFN-adopting economies already sitting on a different,
831 faster-rising emissions trajectory than non-adopters (which a single covariate
832 cannot fully absorb) than with a discrete policy-caused jump at the adop-
833 tion date. And the naive TWFE estimate (0.901) is itself more than double
834 the doubly-robust Callaway-Sant’Anna estimate (0.349), the same direction
835 of bias documented in the staggered-DID literature (Goodman-Bacon, 2021)
836 and a useful internal illustration of why we lead with the more robust esti-
837 mator throughout this section.

838 Both doubly-robust results are stable to the choice of comparison group:
839 substituting not-yet-treated economies for never-treated economies leaves the
840 CO2 estimate essentially unchanged (0.313 versus 0.349, both significant at
841 conventional levels) and the renewable-energy null equally unchanged (0.653

842 versus 0.674, neither distinguishable from zero); Table 8 reports both. Nei-
843 ther conclusion in this section depends on which set of untreated economies
844 serves as the counterfactual.

845 Neither result supports a “green banking policy backfires” reading, and
846 neither supports a “green banking policy works” one. The more measured
847 reading is that a macro-level design – built using exactly the identification
848 strategy (staggered adoption timing, a proper Callaway-Sant’Anna estima-
849 tor, never-treated controls) this literature’s own methodological expectations
850 call for – also does not produce clean evidence that SBFN adoption changes
851 a country’s aggregate environmental trajectory for the better, and the one
852 statistically significant result we do find is more parsimoniously explained by
853 which economies self-select into SBFN membership than by what the policy
854 itself does once adopted. This is, we think, a genuinely different and com-
855plementary piece of evidence to Sections 5.2–5.7’s firm-level results, not a
856 restatement of them: different data source, different unit of analysis, differ-
857ent outcome constructs, and, for the first time in this paper, an identification
858 strategy that exploits exogenous variation in adoption *timing* rather than re-
859lying on cross-sectional comparison across adopters and non-adopters. That
860 it converges on the same substantive conclusion – no clean evidence of a ben-
861eficial institutional effect – from a design built specifically to meet a bar the
862 paper’s firm-level evidence cannot meet, strengthens rather than substitutes
863 for the paper’s central finding.

864 7. Discussion

865 Put in plain economic terms, the paper’s central finding is this: a firm
866 holding a bank credit line is associated with an adoption probability roughly
867 13 percentage points higher than an otherwise-similar firm without one—an
868 association net of observed controls, not a causally identified effect (Limita-
869 tion 1)—and that gap is essentially unchanged whether the firm sits in an
870 SBFN adopter or non-adopter, or across the full range of regulatory qual-
871 ity (Figure 4). Firms in SBFN-adopting economies are, on average, 15–20
872 percentage points *less* likely to have adopted a green practice at all—a level
873 gap, not a slope gap.

874 This convergence is informative precisely because of how differently the
875 evidence could have failed to agree. The primary and global samples share al-
876 most no economies, span different half-decades, use different outcome items,
877 and reach institutional settings—high-income, non-SBFN economies such as
878 Australia and Germany—the primary sample cannot touch. Section 6’s
879 country-level staggered event-study adds a third, independent test on top
880 of this: a different data source, a different unit of analysis, and the paper’s
881 only design that exploits exogenous variation in adoption *timing*. That it
882 too finds no clean evidence of a beneficial institutional effect—a precise null
883 on renewable-energy share, and a CO2 association that shrinks substantially
884 once development stage is controlled for—is the hardest-to-dismiss evidence
885 that this is a real feature of the data, not an artifact of any one sample, level
886 of analysis, or estimator. (This is a distinct claim from H2/H3’s firm-level
887 moderation null—a main effect at country level, not a slope change at firm
888 level—but evidence of the same kind.)

889 This complicates rather than confirms the implicit premise of a green-
890 credit-policy literature disproportionately built on China’s unusually mature,
891 well-resourced guidelines (Li et al., 2023; Huang et al., 2023; Dai et al., 2025).
892 Our sample spans every stage of SBFN’s own progression framework, from
893 Jordan and the Kyrgyz Republic’s initial “Commitment” to Mongolia’s “Ad-
894 vancing,” and across that range we find no evidence that formal adoption
895 changes how credit access translates into green investment. The causal for-
896 est’s feature-importance decomposition (Section 5.4) offers one redirection
897 rather than a tidy answer: the heterogeneity that does exist is overwhelm-
898 ingly a firm-size phenomenon (75.6% of explained variance) rather than in-
899 stitutional (regulatory quality 19.9%, SBFN 2.6%)—though it does not re-
900 solve into a clean size gradient, so we can say where the heterogeneity sits
901 without yet saying in which direction it runs. If the binding constraint is
902 which firms can absorb a green investment’s fixed costs once financed, rather
903 than whether a supervisory guideline nudges banks to lend more broadly, a
904 country-level mandate on the extensive margin of lending may be aimed at
905 the wrong margin regardless of its own credibility.

906 “No interaction detected” is not the same claim as “no institutional com-
907 plementarity exists,” and we do not want the former to be read as settling
908 the latter. Two alternative readings stay live: a linear interaction may be
909 the wrong functional form for a genuinely threshold-shaped complementarity,
910 given SBFN’s own graded progression framework and our binary coding of
911 it (Limitation 2); and the Worldwide Governance Indicators may simply not
912 proxy the banking-supervisory capacity a green-lending guideline depends
913 on (Limitation 5). We think our preferred reading—no complementarity

914 through this channel—is best supported by the evidence, but hold it as the
915 best current account rather than a claim that rules the alternatives out.

916 Two further limits are worth naming directly. First, SBFN guidelines
917 could in principle be working on the *extensive* margin of who obtains credit
918 at all, rather than the *intensive* margin our interaction terms test; testing
919 that would require modeling credit access itself as a function of SBFN status,
920 which our demand-side design does not do. Second, every outcome in this
921 paper is firm-reported, and we cannot rule out a compliance-reporting-versus-
922 practice gap between what firms report and what banks report to their own
923 supervisor—a limit no firm-level survey instrument can fully resolve on its
924 own. Read together with Section 6, whose null is, to our knowledge, the first
925 test of the SBFN framework’s own theory of change against outcomes on
926 the ground rather than its own tracked policy-stage progression, both point
927 the same way: a framework can look like it is working by every measure the
928 network itself tracks while the channel it is meant to move remains, on this
929 evidence, unmoved.

930 The one interaction that clears a conventional significance threshold—a
931 negative credit $\times$ SBFN effect for overdraft rather than term-credit access
932 (Section 5)—is more consistent with this redirection than with a simple policy
933 failure: overdrafts finance working capital, not the long-horizon expenditure
934 our outcome measures, so a supervisor successfully steering credit toward
935 term instruments would produce exactly this pattern. We flag it as suggestive
936 only: it is one significant coefficient among many tested, and does not survive
937 a Benjamini–Hochberg correction at any conventional false discovery rate
938 (Supplementary Table S3).

939 Read against the broader themes of state capacity and policy credibility,
940 our results sit closer to a state-capacity-is-necessary-but-firm-level-frictions-
941 bind-independently story than a straightforward complementarity one. Reg-
942 ulatory quality raises the baseline propensity to adopt green practices, but
943 is not the missing ingredient that lets a green banking guideline reshape
944 the credit-to-green channel specifically. The binding constraint looks less
945 like whether the supervising jurisdiction can make its guideline credible, and
946 more like whether firms below a certain scale can absorb a green investment’s
947 fixed costs once credit is available—a question one level down from the one
948 the SBFN framework is designed to answer.

949 8. Conclusion

950 This paper set out to test whether the wave of green banking policy
951 adoption coordinated through the Sustainable Banking and Finance Net-
952 work changes the relationship between firm-level access to finance and green
953 investment behavior, and whether that depends on the regulatory capacity
954 available to make the policy credible. Using harmonized WBES data across
955 41 economies (2018–2020), a four-country waste-minimization check, and an
956 independently assembled 162-economy global sample (2021–2026) reaching
957 for the first time into high-income, non-SBFN economies, we apply four es-
958 calating estimators and find five consistent results. First, bank credit access
959 is robustly associated with green practice adoption, surviving every specifi-
960 cation, subsample, and alternative-outcome check, and replicating in both
961 the small waste-minimization sample and the full global sample. Second,
962 neither SBFN adoption nor regulatory quality moderates that relationship

963 under any estimator, on either sample, though both shift the general level of
964 adoption directly in the primary sample. Third, where effect heterogeneity
965 does exist, it sits at the level of firm size rather than country institutions.
966 Fourth, this pattern replicates on a sample sharing almost no economies with
967 the primary one, under a materially sharper identification strategy. Fifth,
968 and most importantly for the paper’s credibility, an independent country-
969 level staggered event-study (Section 6)—a main-effect test, not a repeat of
970 the firm-level moderation null—likewise finds no clean evidence that SBFN
971 adoption improves aggregate environmental outcomes, the single strongest
972 evidence that this null is a real feature of the data rather than an artifact of
973 any one sample, data source, level of analysis, or estimator.

974 For policy, the implication is not that green banking guidelines are in-
975 effective, but that our evidence does not support treating them as a lever
976 for widening the green-investment gap between financed and unfinanced
977 firms—the margin most of this literature implicitly assumes they operate
978 on. If our tentative loan-tenor reading of the overdraft result is correct,
979 policymakers may find more traction focusing on the term-lending instru-
980 ments that actually finance green capital expenditure, and on complementary
981 measures—technical assistance, matching grants, credit guarantees—that ad-
982 dress smaller firms’ absorptive-capacity constraints directly.

983 We close with seven limitations. First, both firm-level samples are sin-
984 gle cross-sections (Section 4 explains why a genuine panel does not exist for
985 this outcome): they cannot rule out reverse causality or unobserved coun-
986 try characteristics correlated with both SBFN adoption and baseline green-
987 adoption propensity, and the global sample’s fixed effects sharpen this only

988 for the interaction terms, not the credit main effect. Second, SBFN mem-
 989 bership is a binary proxy for what the network’s own framework treats as
 990 a graded process; the global sample’s regional roster entries (the CEMAC
 991 and Eastern Caribbean groupings, Section 4) additionally rest on an infer-
 992 ence about coverage rather than an explicit per-country statement. Third,
 993 the four-economy waste-minimization check cannot re-test the SBFN con-
 994 trast (three of its four economies are members) and combines non-identical
 995 outcome items by construction of WBES’s own shortened module design—a
 996 data-availability constraint, not a methodological choice. Fourth, the global
 997 sample’s CO2-monitoring outcome is a single item, narrower than the pri-
 998 mary sample’s seven-item composite; that the same pattern holds on it is
 999 reassuring but does not establish it would hold on a richer measure. Fifth,
 1000 the Worldwide Governance Indicators measure is a broad governance indica-
 1001 tor, not a banking-supervision-specific capacity measure. Sixth, our design
 1002 tests only the intensive margin of the credit-to-green channel and cannot
 1003 rule out that SBFN policy operates on the extensive margin of who obtains
 1004 credit in the first place—a margin our realized-access measure cannot ob-
 1005 serve, since some firms decline formal credit altogether (Jin, 2026)—nor dis-
 1006 tinguish firm-reported practice adoption from a bank-compliance-reporting
 1007 gap (Section 7). Seventh, Section 6’s event-study is an ecological-level test
 1008 of a related but distinct question; its one significant result (CO2 per capita)
 1009 is netted against a single covariate, and we cannot rule out that a richer set
 1010 of development-stage controls would shrink it further, or that some genuine,
 1011 small, confounded relationship remains. We leave each of these to future
 1012 work.

1013 **Declaration of generative AI and AI-assisted technologies in the**
1014 **manuscript preparation process**

1015 During the preparation of this work the author(s) used AI-assisted tools
1016 to help polish the writing. After using these tools, the author(s) reviewed
1017 and edited the content as needed and take full responsibility for the content
1018 of the published article.

1019 **Supplementary material**

1020 The full per-economy SBFN membership and regulatory-quality cod-
1021 ing for both samples, Bayesian convergence diagnostics, the global sam-
1022 ple's SBFN membership by region, and the additional robustness checks and
1023 multiple-testing correction referenced above (Section 5.5, 5.6) are reported
1024 in the Supplementary Material accompanying this submission.

1025 *Data availability*

1026 Data in support of the findings of this study is available from the cor-
1027 responding author upon reasonable request, subject to the World Bank En-
1028 terprise Surveys' own terms of use for the underlying firm-level microdata
1029 (enterprisesurveys.org, free registration required), including the standard-
1030 ized cross-country database and Productivity Database underlying the global
1031 sample. The country-level SBFN policy-status (including the SBFN data
1032 portal's member roster underlying the global sample) and Worldwide Gover-
1033 nance Indicators datasets constructed for this paper, the country-year World
1034 Development Indicators panel and Callaway-Sant'Anna event-study code un-
1035 derlying Section 6 (all freely redistributable, no registration required), to-

1036 together with the full analysis and replication code, are available at [https:](https://github.com/vinhqdang/green-credit-substitution)
1037 [//github.com/vinhqdang/green-credit-substitution](https://github.com/vinhqdang/green-credit-substitution).

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